NEF ABUNDANCE: FRAMEWORK – FAILURE OF JET SEPARATION

1. Failure of Jet Separation

Let X be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor on X such that $\mathrm{n}(X, D) = 3$ and $D^3 = 0$. Assume in addition that $\kappa(D) = -\infty$.

Setup.

- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small such that
$$343\epsilon^3 < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$
- Fix an ample Cartier divisor A. For integers $n \geq 1$, put
$$a_n = ((D + A/n)^3)^{1/3}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$
Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^3 \to \epsilon^3$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large n.
- Let $p_i : X \times X \to X$ be the projections, for $i = 1, 2$. Put
$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big).$$
Let $\pi : Z \to X \times X$ be the projection. Let $S_1, S_2 \subset Z$ be the two summand sections and set
$$U = Z \setminus (S_1 \cup S_2).$$
- Set $\xi = c_1(\mathcal{O}_Z(1))$. For a positive integer s, put
$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$
Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with $kR_{n,s}$ Cartier, equivalently kP_n Cartier.

Jet separation. Let W be a smooth variety, L a Cartier divisor, and $z \in W$. For an integer $j \geq 0$, $|L|$ separates j-jets at z if
$$\forall u \in \mathcal{O}_{W,z}(L) \quad \exists \sigma \in H^0(W, \mathcal{O}_W(L)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1}\mathcal{O}_{W,z}(L).$$
Here $\mathfrak{m}_z \subset \mathcal{O}_{W,z}$ is the maximal ideal.

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Lemma 1.2. *For every positive integer s there exists an integer $n_0 = n_0(\epsilon, s) \geq 1$ such that, for every integer $n \geq n_0$, $|kR_{n,s}|$ does not separate $8k$-jets at any $z \in U$, for any positive integer k with $kR_{n,s}$ Cartier.*

Draft status. The proof below is extracted from the existing threefold proof draft. Its source-dependent inputs and arguments have not been independently verified. This note isolates the indicated framework lemma for discussion and checking.

2. Proof of Lemma 1.2

2.1. The scalar input.

Lemma 2.1. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$, $D^3 = 0$, and $\kappa(D) = -\infty$. Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$ such that $343\epsilon^3 < 1/4$ and $40\epsilon < 3/8$. For $n \geq 1$, let r_n be the nearest positive integer to $\epsilon/((D + A/n)^3)^{1/3}$ and set $P_n = r_n(D + A/n)$. For every integer $s > 0$ and all $n \gg 0$, let $\beta : \widetilde{X} \to X$ be the blowup at a very general point x, with exceptional divisor E. There are a birational morphism $f : V \to \widetilde{X}$ from a smooth projective threefold and ample Cartier divisors H_1, H_2 on V such that*

$$f^*\beta^*(P_n + sD) \cdot H_1 \cdot H_2 < 40\epsilon\, f^*E \cdot H_1 \cdot H_2.$$

Write $\nu(D) = \max\{j : D^j \cdot A^{3-j} > 0\} \in \{1, 2\}$. We obtain the curve bound from a multiplicity estimate and duality. All auxiliary varieties in this section are over $\mathbb{C}$.

2.1.1. *Normalization and numerical inputs.*

Lemma 2.2. *For every $\epsilon > 0$ and as $n \to \infty$ through the positive integers, there are positive integers $r_n \to \infty$ such that*

$$P_n = r_n(D + A/n), \qquad P_n^3 \longrightarrow \epsilon^3.$$

The divisor P_n is ample and rational Cartier; its Cartier multiples are chosen after this normalization.

Proof. Put $a_n = ((D + A/n)^3)^{1/3}$. Then $a_n > 0$ and $a_n \to 0$. Choose r_n to be the nearest positive integer to ϵ/a_n. For sufficiently large n, $|r_n a_n - \epsilon| \leq a_n/2$, which proves the assertions. □

Lemma 2.3. *Let (Y, B) be an auxiliary projective klt pair of dimension $f \in \{1, 2\}$, with $T = K_Y + B$ pseudo-effective, and let M be nef and rational Cartier. Then $T + M$ has an effective rational numerical representative. Moreover, if for some $a > 0$*

$$(T + aM) \cdot C > 0$$

for every integral curve through a sufficiently very general point, then $T + cM$ is big for every rational $c > 0$.

Proof. Numerical nonvanishing on surfaces is [4, Corollary C]; on curves it follows from the degree criterion. For the second assertion, the curve case is immediate. In dimension two run the ordinary $(K_Y + B)$-minimal model program, after a small $\mathbb{Q}$-factorialization. Its output is a morphism $g : Y \to Y_0$ with $U = K_{Y_0} + B_0$ nef. The pushforward $M' = g_*M$ is nef. Negativity gives

$$T = g^*U + E_T, \qquad M = g^*M' - E_M,$$

where $E_T, E_M \geq 0$ are exceptional; E_T has positive coefficient on every contracted divisor. Thus, for small rational $t > 0$, $E_T - tE_M \geq 0$.

By [4, Corollary C], $U+tM'$ is numerically equivalent to a semiample rational divisor. If it were not big, the resulting morphism would have positive-dimensional fibres. A general fibre, or a general curve in such a fibre when the image is a point, avoids the finitely many points $g(\operatorname{Exc} g)$. Since U and M' are nef, this curve has both U- and M'-degree zero. Its inverse image contradicts the stated full positivity of $T+aM$. Consequently $U+tM'$, and hence $T + tM = g^*(U + tM') + E_T - tE_M$, is big. For $c > 0$, choose $0 < \lambda < \min\{1, c/t\}$ and write

$$T + cM = \lambda(T + tM) + (1 - \lambda)T + (c - \lambda t)M.$$

The last two summands are pseudo-effective, so the sum is big. This proves exactly the dimension-one and dimension-two inputs used in [3, Sections 3–5], without assuming threefold nef nonvanishing. □

2.1.2. *Moving base components.*

Lemma 2.4. *Let W be smooth projective of dimension d, let H be a nef Cartier divisor on W, and let $\Lambda \subseteq |H|$ be a nonempty linear system with base ideal $\mathfrak{b}$. Let V be an integral component of its base locus, of codimension $c \geq 1$. If*

$$\mathfrak{b}\mathcal{O}_{W,\eta_V} \subseteq \mathfrak{m}^h_{\eta_V} \qquad (h \in \mathbb{Z}_{>0}),$$

then

$$h^c\, H^{d-c} \cdot V \leq H^d.$$

In particular, if x is an isolated base point and every member has multiplicity at least a real number $r \geq 0$ at x, then $r^d \leq H^d$.

Proof. Intersect successively with general members of Λ. Before each next cut, remove the components already contained in the base locus. Each remaining component is cut properly, and the intersection cycle is effective. Its H-degree equals the H-degree of the

preceding cycle; discarding effective components can only decrease this degree, since H is nef. Starting with $[W]$, the H-degree of the codimension-c cycle is therefore at most H^d.

At η_V, the localized base ideal is primary for the maximal ideal. No discarded component of smaller codimension contains η_V, because V is an irreducible component of the base locus. The first c general equations $s_1, \dots, s_c$ form a parameter ideal I in the regular local ring $R = \mathcal{O}_{W,\eta_V}$. The coefficient of V in the intersection cycle is $\operatorname{length}(R/I)$. Since $I \subseteq \mathfrak{m}^h$, monotonicity of Hilbert–Samuel multiplicity gives

$$\operatorname{length}(R/I) = e(I, R) \geq e(\mathfrak{m}^h, R) = h^c.$$

Here the first equality uses that a parameter sequence in R is regular, and the last equality uses regularity of R. This proves the inequality. For an isolated point take $c = d$ and $h = \max\{1, \lceil r \rceil\}$. □

Lemma 2.5. *Let W be smooth projective, $\Theta \geq 0$ rational, and $V \subset W$ integral of positive dimension. Suppose (W, Θ) is lc at η_V and has an lc place centred on V. For $L = (K_W + \Theta)|_{V^\nu}$ there exist a finite surjective morphism $\eta : V' \to V^\nu$, with V' normal, and rational divisors $B_{V'} \geq 0$ and $M_{V'}$, the trace of a b-nef divisor, such that*

$$\eta^* L \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'}.$$

Proof. We use the construction in [1, Lemma 7.1]. The dlt blowup [2, Theorem 2.10] gives

$$p^*(K_W + \Theta) = K_Q + B_Q + E_Q, \qquad (Q, B_Q) \text{ dlt}, \quad E_Q \geq 0,$$

where B_Q is the truncated boundary and the effective surplus E_Q is absent near η_V. Choose a dlt stratum S minimal among those dominating V. Adjunction, followed by adding $E_Q|_S$, gives an effective rational divisor B_S and

$$K_S + B_S \sim_{\mathbb{Q}} (p|_S)^*((K_W + \Theta)|_V).$$

The stratum is not contained in E_Q. Minimality makes (S, B_S) klt over η_V, although it need not be globally lc.

Factor through the normalization and the connected-fibre base: $S \xrightarrow{q} V' \xrightarrow{\eta} V^\nu \to V$, with η finite. The discrepancy-round-up rank condition for q is one: over the generic point the effective klt boundary contributes zero round-up on strict divisors and allows poles only along exceptional divisors. Its pushforward to normal S is therefore $\mathcal{O}_S$, and connectedness gives rank one. Thus this is a klt-trivial fibration in [5, Definition 3.1]. Its formula is

$$\eta^*((K_W + \Theta)|_{V^\nu}) \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'},$$

with b-nef moduli divisor M by [5, Theorem 3.5]. The discriminant on this initial base is effective: over a prime divisor, an inverse-image component of multiplicity $m \geq 1$ and boundary coefficient $b \geq 0$ bounds its lc threshold by $(1-b)/m \leq 1$. A negative threshold still gives a positive discriminant coefficient. Only generic sub-klt singularities are required for this use of the formula. □

Lemma 2.6. *Under the hypotheses of Lemma 2.5, let $f = \dim V$, let P_V be nef and rational Cartier, and let $\rho : Y \to V$ be a resolution. Then*

$$K_Y \cdot (\rho^* P_V)^{f-1} \leq (K_W + \Theta)|_V \cdot P_V^{f-1}.$$

Proof. Apply Lemma 2.5 and intersect with $f-1$ copies of the pulled-back P_V. The effective discriminant and the moduli trace have nonnegative intersection; for the latter, push down from a higher base where the moduli divisor is nef. For compatible canonical Weil cycles,

$$\eta_* K_{V'} = (\deg \eta) K_{V^\nu} + \eta_* R_\eta, \qquad R_\eta \geq 0.$$

Discard ramification and divide by $\deg \eta$. Pushing down K_Y to V^ν changes it only by exceptional cycles, whose testing intersection is zero. No rational Cartier assumption on K_{V^ν} is needed. □

Lemma 2.7. *Let W be smooth projective of any dimension, and let P_W be nef and rational Cartier. Suppose a positive-dimensional integral base component V of a subseries of $|kP_W|$ is isolated at its generic point, of codimension a, and its base ideal there is contained in $\mathfrak{m}^h$. Then on a resolution $\rho : Y \to V$,*

$$K_Y \cdot (\rho^* P_W)^{\dim V - 1} \leq (K_W + cP_W)|_V \cdot P_W^{\dim V - 1}, \qquad 0 < c \leq ak/h.$$

Proof. The generic log canonical threshold of the base ideal satisfies $\lambda \leq a/h$, as seen by blowing up the smooth generic centre. On a log resolution, average sufficiently many general members of the subseries with total coefficient λ. This gives an actual effective boundary $\Theta \sim_{\mathbb{Q}} k\lambda P_W$ which is lc at the generic point of V and has an lc place centred on V. Lemma 2.6 gives the inequality with $c = k\lambda$. The construction uses actual general members of the given subseries, and requires lc only at η_V. Both the threshold bound and subadjunction are dimension-independent. □

Lemma 2.8. *Let H be a Cartier divisor on X, let $\Lambda \subseteq |H|$ be a fixed nonempty linear system, and let $r \geq 0$ be real. If a Zariski dense set of points x admits $E_x \in \Lambda$ with $\operatorname{mult}_x(E_x) > r$, then every point does.*

Proof. Use local parameters u, v, w and homogeneous parameters $[t_0 : \cdots : t_m]$ on the fixed linear system. Write the universal equation as $f = \sum_i t_i f_i$, and denote its coefficient

functions by $c_{i,\alpha}(x)$, where $\alpha \in \mathbb{Z}^3_{\geq 0}$. The incidence Z is defined locally by

$$\sum_i t_i c_{i,\alpha}(x) = 0 \qquad (|\alpha| \leq \lfloor r \rfloor).$$

These coefficients are the finite local jet coefficients. Thus Z is closed, and its proper projection has closed image. An image containing a Zariski dense subset equals X.

□

Lemma 2.9. *Let W be smooth projective, L a line bundle, and $z : T \to W$ a morphism from a smooth integral variety. Fix integers $m_1 > m_0 \geq 0$ and $1 \leq h \leq m_1 - m_0 + 1$. Suppose the spaces*

$$\mathcal{H}_i(t) = H^0(W, L \otimes \mathfrak{m}_{z(t)}^{m_i}) \quad (i = 0, 1)$$

form vector bundles, and a reduced family $\Gamma \subset T \times W$ dominates W, with each general fibre Γ_t a component of both base loci. Then the higher base ideal satisfies $\mathfrak{b}_{1,t} \subset I^h_{\Gamma_t}$ at a general incidence point.

Proof. A frame of the higher kernel has the form $s(t, w) = \sum_\alpha a_\alpha(t) s_\alpha(w)$ for fixed global sections. A parameter derivative of order $j < h$ loses at most j orders at the moving mark, so belongs to the full lower kernel and vanishes on Γ. At a general smooth point, dominance of $\Gamma \to W$ makes parameter directions span the normal space. Choose transverse parameters t' and write Γ locally as $u = t' - F(t'', w) = 0$. Differentiation in t' with t'', w fixed is differentiation in u. Thus all terms of degree less than h in the normal parameters vanish, giving $s \in I^h_\Gamma$. Characteristic zero and faithful flatness of the analytic local ring give the algebraic containment. Restricting the finite frame to a general parameter fibre proves the assertion, as in [1, Lemma 7.3]. □

Lemma 2.10. *Let $g : W \to B$ be a morphism from a smooth projective variety, and let an integral component V of the base locus of a linear system have base ideal $\mathfrak{b} \subset I^h_V$. Let F be the reduced component through a general point of a fibre of $V \to g(V)$. Assume the ambient fibre W_b is smooth and integral and $F \subsetneq W_b$. Then the restricted system is nonzero, F is a base component, and*

$$\mathfrak{b}|_{W_b} \subset I^h_F$$

near η_F.

Proof. Remove the other base components and work where $V \to g(V)$ is smooth over its actual image. Its scheme fibre is reduced there, hence $I_V \mathcal{O}_{W_b} = I_F$ and $I^h_V \mathcal{O}_{W_b} = I^h_F$. The local base support is exactly F. Since F is proper, not all restricted sections vanish identically on W_b. The resulting base ideal is primary at η_F. This is the actual-image restriction in [1, Lemma 7.4]. □

Lemma 2.11. *Let W be a smooth projective variety of dimension d, and let P_W be ample and rational Cartier. Fix kP_W Cartier and levels $\tau_0 < \cdots < \tau_d$ with common gap $\delta > 0$ and $k\delta \geq 4$. Suppose that, at a very general marked point z, all full spaces*

$$\mathcal{H}_i(z) = H^0(W, \mathcal{O}_W(kP_W) \otimes \mathfrak{m}_z^{\lceil k\tau_i \rceil})$$

are nonzero, and the lowest base locus has a positive-dimensional component through z. Then there is an algebraic covering family of proper integral components V_t, common to two adjacent base loci, with $1 \leq f = \dim V_t < d$. At a general incidence point the entire higher-kernel base ideal satisfies

$$\mathfrak{b}_t \subset I_{V_t}^h, \qquad h = \lceil k\delta/2 \rceil, \qquad k/h \leq 2/\delta.$$

Consequently, on a resolution V_t^*,

$$\begin{aligned} &0 < P_W^f \cdot V_t \leq (2/\delta)^{d-f} P_W^d, \\ &K_{V_t^*} \cdot P_W^{f-1} \leq (K_W + cP_W)|_{V_t} \cdot P_W^{f-1}, \qquad 0 < c \leq 2(d-f)/\delta. \end{aligned} \tag{1}$$

Proof. On the constant-rank open, the full kernels are vector bundles with nested proper base loci. Track a component through the mark across the $d+1$ levels. Its dimensions lie in $\{1, \ldots, d-1\}$, so two adjacent tracked components coincide. After a finite extension of the parameter field, spread them out to an integral family whose marked incidence dominates W. The adjacent orders satisfy

$$\lceil k\tau_{i+1} \rceil - \lceil k\tau_i \rceil \geq k\delta - 1 \geq h - 1.$$

Lemma 2.9 gives the ordinary-power containment for the entire higher base ideal. Apply Lemmas 2.4 and 2.7 for the degree and canonical bounds. The construction follows [3, Lemmas 4.5–4.6]. □

Proposition 2.12. *Let P be an ample rational Cartier divisor on X, and put $\rho = (P^3)^{1/3}$. Suppose that, for some positive integer k_* with k_*P Cartier, a Zariski dense set of points x admits $E_x \in |k_*P|$ with $\operatorname{mult}_x(E_x) > 4k_*\rho$. Then there is a covering family of integral subvarieties V of one fixed dimension $d \in \{1, 2\}$ such that*

$$0 < P^d \cdot V \leq 3^{3-d}(P^3)^{d/3}.$$

Moreover, on a resolution $Y \to V$, with P_Y the pullback,

$$K_Y \cdot P_Y^{d-1} \leq \frac{3(3-d)}{\rho} P_Y^d. \tag{2}$$

The marked incidence dominates X.

Proof. Lemma 2.8 makes the multiplicity condition valid at every point. Taking powers of the defining sections allows k to be arbitrarily large and divisible by k_*. Put

$$\delta = \frac{2\rho}{3}, \qquad \tau_i = \frac{3}{2}\rho + i\delta \quad (0 \le i \le 3), \qquad q_i = \lceil k\tau_i \rceil.$$

Choose k so that $k\delta \ge 4$ and $k\rho/2 > 1$. All full kernel spaces

$$V_i(x) = H^0(X, \mathcal{O}_X(kP) \otimes \mathfrak{m}_x^{q_i})$$

are nonzero, because $\tau_3 = 7\rho/2 < 4\rho$. Write $B_i(x)$ for their base loci. They are proper and nested. The point x cannot be isolated in $B_0(x)$: otherwise Lemma 2.4 would give $q_0^3 \le k^3P^3$, whereas $q_0 \ge 3k\rho/2$.

Apply Lemma 2.11 with $W = X$ and these four levels. It gives a common component V_t of dimension $d \in \{1, 2\}$ and a covering incidence, together with ordinary-power containment for the entire higher base ideal, where $h = \lceil k\delta/2 \rceil$.

The higher base ideal is primary there, since V_t is its component. Lemma 2.4, applied to $H = kP$, gives

$$h^{3-d}k^dP^d \cdot V_t \le k^3P^3, \qquad P^d \cdot V_t \le (k/h)^{3-d}P^3 \le (3/\rho)^{3-d}P^3.$$

Lemma 2.7 gives $K_Y \cdot P_Y^{d-1} \le (3-d)(k/h)P_Y^d$, because $K_X \sim 0$. Since $k/h \le 3/\rho$, this is (2). Ampleness gives positivity, and the incidence is dominant. □

2.1.3. *The scalar bound and the movable class.*

Lemma 2.13. *Let Y be an auxiliary smooth projective variety of dimension $f \in \{1, 2, 3\}$. Let B_0 be a reduced simple normal crossing divisor or zero, L a nef Cartier divisor of full nef dimension, and P_Y a nef rational Cartier divisor. If*

$$0 < P_Y^f \le C_0, \quad P_Y - rL \text{ is nef}, \quad K_Y + B_0 + L \text{ is big},$$

and

$$(K_Y + B_0 + L) \cdot P_Y^{f-1} \le C_1 P_Y^f,$$

then

$$r \le (m_f C_1)^{f-1} C_0,$$

where m_f depends only on f.

Proof. Birkar–Zhang [6, Theorem 1.3], with boundary coefficients in $\{0, 1\}$ and Cartier index one for L, makes $|m_f(K_Y + B_0 + L)|$ birational. Resolve its base ideal and let H be the free moving part. Then H is nef and big and

$$H \cdot P_Y^{f-1} \le m_f C_1 P_Y^f.$$

Khovanskii–Teissier gives

$$P_Y \cdot H^{f-1} \le P_Y^f \left(\frac{H \cdot P_Y^{f-1}}{P_Y^f} \right)^{f-1} \le (m_f C_1)^{f-1} C_0.$$

General hyperplanes through the image of a very general point under the birational morphism defined by H cut an effective curve cycle containing a curve C through that point. Full nef dimension and integrality give $L \cdot C \ge 1$. Thus $r \le P_Y \cdot C \le P_Y \cdot H^{f-1}$. For $f = 1$ take $C = Y$. This is [3, Lemma 4.1]; in particular its constants are independent of the varying auxiliary variety. □

Proposition 2.14. *Assume* $\mathrm{n}(X, D) = 3$ *and* $D^3 = 0$. *With* $P_n = r_n(D + A/n)$ *as in Lemma 2.2, put* $\rho_n = (P_n^3)^{1/3}$. *For all sufficiently large* n, *and at a very general point* x, *every member* $E \in |kP_n|$, *for every positive integer* k *with* kP_n *Cartier, satisfies*

$$\mathrm{mult}_x(E) \le 4k\rho_n.$$

Proof. Suppose violations persist as $n \to \infty$. For fixed n, k, the high-multiplicity condition is closed in the incidence space, so a violation at a sufficiently very general point yields a dense set of violations. Use Proposition 2.12, together with its canonical bound (2), and write $Y \to V$ for the resolution of a general member. For $f = \dim V \in \{1, 2\}$ it gives

$$0 < P_Y^f \le 3^{3-f} \rho_n^f, \qquad K_Y \cdot P_Y^{f-1} \le \frac{3(3-f)}{\rho_n} P_Y^f.$$

The volume window $P_n^3 \to \epsilon^3 > 0$ makes both bounds uniform in n. Let L be the pullback of D; then $P_Y - r_n L$ is nef.

Choose the incidence point above the very general full-nef locus and outside the images of all families of rational curves. The latter exclusion is possible because $K_X \sim 0$ makes X non-uniruled. Since $Y \to V \subset X$ is generically finite, Y is non-uniruled and L has full nef dimension; this is the very-general-images argument of [3, Lemma 4.2]. By [7], K_Y is pseudo-effective, and Lemma 2.3 makes $K_Y + L$ big. Also $L \cdot P_Y^{f-1} \le P_Y^f / r_n$. Lemma 2.13 now bounds r_n uniformly, contrary to $r_n \to \infty$. There are only countably many n, k and jet orders, so one very general locus handles all the stated section spaces. □

Proposition 2.15. *Assume the scalar bound at a very general point* x, *and fix large* n *with* $\rho_n < 2\epsilon$ *and* $r_n > s$. *Let* $\beta : \widetilde{X} \to X$ *be the blowup at* x, *with exceptional divisor* E. *There are a birational morphism* $f : V \to \widetilde{X}$ *from a smooth projective threefold and ample Cartier divisors* H_1, H_2 *on* V *such that*

$$f^*\beta^*(P + \lambda D) \cdot H_1 \cdot H_2 < 40\epsilon\, f^*E \cdot H_1 \cdot H_2 \quad (0 \le \lambda \le s).$$

Proof. The class $2\beta^*P - 40\epsilon E$ is not pseudo-effective. Otherwise its convex combination with the big class $2\beta^*P$ would make $2\beta^*P - cJ$ big for rational $8\rho_n < c < 40\epsilon$. An effective divisor in a divisible multiple of this class would push down to a member of $|2kP|$ with multiplicity exceeding the bound in Proposition 2.14. Movable-curve duality [7] gives a smooth projective birational model $f : V \to \widetilde{X}$ and ample real divisor classes H_1, H_2 such that

$$2f^*\beta^*P \cdot H_1 \cdot H_2 < 40\epsilon\, f^*E \cdot H_1 \cdot H_2.$$

Strict negativity permits rational approximation and clearing denominators, so the H_i may be taken ample Cartier. For $0 \le \lambda \le s$ the divisor

$$2P - (P + \lambda D) = (r_n - \lambda)D + r_n A/n$$

is nef, which gives the required inequalities. Fix this actual model and complete intersection before reducing characteristic. □

Proof of Lemma 2.1. Fix s and take n sufficiently large that Proposition 2.14 applies, $\rho_n < 2\epsilon$, and $r_n > s$. Apply Proposition 2.15 with $P = P_n$ and $\lambda = s$. □

2.2. **The Frobenius input.**

Lemma 2.16. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$ and $D^3 = 0$. Let P be an ample rational Cartier divisor and $s > 0$ an integer such that*

$$343P^3 < \tfrac{1}{4}, \qquad (2P + sD) \cdot P^2 < 4P^3.$$

Let $\beta : \widetilde{X} \to X$ be the blowup at a point x, with exceptional divisor E. Suppose that there are a birational morphism $f : V \to \widetilde{X}$ from a smooth projective threefold and ample Cartier divisors H_1, H_2 on V such that

$$f^*\beta^*(P + sD) \cdot H_1 \cdot H_2 < \tfrac{3}{8} f^*E \cdot H_1 \cdot H_2.$$

Let D_i, P_i denote pullbacks from the i-th factor of $X \times X$ and then to

$$Z = \mathbb{P}_{X\times X}\big(\mathcal{O}_{X\times X}(D_1) \oplus \mathcal{O}_{X\times X}(D_2)\big), \qquad \xi = c_1(\mathcal{O}_Z(1)).$$

Let $\pi : Z \to X \times X$ be the bundle projection, and write $D_i = (p_i \circ \pi)^*D$, $P_i = (p_i \circ \pi)^*P$ on Z. Put $R(P, s) = P_1 + P_2 + s\xi$, and let U be the complement of the two summand sections. For every integer $k > 0$ with kP Cartier, $|kR(P, s)|$ does not separate $8k$-jets at any point of U.

Throughout this subsection, $X, D, P, s, x, \beta, E, f, V, H_1, H_2$ satisfy the hypotheses of Lemma 2.16. Since f^*E is nonzero and effective and H_1, H_2 are ample, $f^*E \cdot H_1 \cdot H_2 > 0$. Put

$$b = \frac{f^*\beta^*(P + sD) \cdot H_1 \cdot H_2}{f^*E \cdot H_1 \cdot H_2} < \tfrac{3}{8}.$$

Nefness of D gives, for $0 \le \lambda \le s$,

$$f^*\beta^*(P + \lambda D) \cdot H_1 \cdot H_2 \le b\, f^*E \cdot H_1 \cdot H_2.$$

Suppose for contradiction that $|k_0 R(P, s)|$ separates $8k_0$-jets at some $z \in U$, for an integer $k_0 > 0$ with $k_0 P$ Cartier. All these data are fixed over $\mathbb{C}$ before reduction. The following estimates specialize [3, Theorem 6.1]; Frobenius is used only on smooth positive-characteristic fibres.

2.2.1. *Fixed data and generic rank.*

Lemma 2.17. *For the fixed ample rational Cartier divisor P, there exist positive integers l_1, l_2, with $l_i P$ Cartier, and divisors $S \in |l_1 P|$, $S_2 \in |l_2 P|$ such that $X \supset S \supset C = S \cap S_2$ is a smooth integral complete intersection flag and $\Omega^1_X|_C$ is semistable of degree zero. In particular, $\mu_{\min}(\Omega^1_X|_C) = 0$.*

Proof. Choose an integer $a > 0$ such that $H = aP$ is very ample Cartier. Since $K_X \sim 0$, generic cotangent semipositivity [8, Theorem 2.1] gives $c_1(\mathcal{Q}) \cdot H^2 \ge 0$ for every torsion-free quotient $\mathcal{Q}$ of Ω^1_X. For a nonzero proper saturated subsheaf $\mathcal{F} \subset \Omega^1_X$, put $\mathcal{Q} = \Omega^1_X/\mathcal{F}$. Then

$$\mu_H(\mathcal{F}) = -\frac{c_1(\mathcal{Q}) \cdot H^2}{\operatorname{rk} \mathcal{F}} \le 0 = \mu_H(\Omega^1_X).$$

Thus Ω^1_X is H-semistable of slope zero.

Mehta–Ramanathan restriction [9, Theorem 6.1] provides sufficiently large integers m_1, m_2 and general divisors $S \in |m_1 H|$, $S_2 \in |m_2 H|$ for which $\Omega^1_X|_{S \cap S_2}$ is semistable. Bertini ensures that S and $C = S \cap S_2$ are smooth and integral. Set $l_i = a m_i$. Finally,

$$\deg(\Omega^1_X|_C) = K_X \cdot C = 0, \qquad \mu_{\min}(\Omega^1_X|_C) = 0.$$

The same construction uses $\dim X - 1$ general high-degree cuts in any dimension when $K_X \equiv 0$. The flag is chosen over $\mathbb{C}$ and fixed before the reduction in Lemma 2.20. □

Lemma 2.18. *If sections of a line bundle L on Z generate jets through order j at z, then products of ℓ such sections generate jets through order ℓj for L^ℓ.*

Proof. Every monomial of degree at most ℓj is a product of ℓ monomials of degrees at most j. Lift the latter to sections with those leading terms and multiply. The product has the desired leading term and only higher-order errors. In a basis ordered by total

degree this gives a triangular matrix with nonzero diagonal on the finite jet quotient, hence surjectivity. □

Proposition 2.19. *Let Y be a smooth positive-characteristic fibre of a model of X, and use the same notation for the specialized divisor data, over an algebraically closed field of characteristic p. On $\mathbb{P}(\mathcal{O}_Y(D)_1 \oplus \mathcal{O}_Y(D)_2)$, suppose a fixed finite system of sections of $k_0(P_1 + P_2 + s\xi)$ generates jets through order $8k_0$ at a point z off the axes, where k_0P is Cartier. Choose T so that all $T + aD$ for $0 \le a \le (k_0 - 1)s$ have sections nonzero at the two base points. Put $\ell_p = \lfloor p/k_0 \rfloor$, $B_p = k_0\ell_p P + T$. For $p \ge 8k_0$, the two-slot evaluation map has generic rank p^6.*

Proof. Lemma 2.18 gives order $8k_0\ell_p$ for $\ell_p k_0 R$. Put $d_p = (p - k_0\ell_p)s$. A weight-zero section of $d_p\xi + T_1 + T_2$ nonzero at z pads this system to $pN\xi + B_{p,1} + B_{p,2}$. Its multiplication is invertible on the local jet quotients. Moreover

$$8k_0\ell_p \ge 8(p - k_0 + 1) \ge 7(p - 1).$$

In seven regular parameters, a monomial surviving modulo $(u_1^p, \dots, u_7^p)$ has degree at most $7(p-1)$. Thus the padded sections surject onto this Frobenius quotient.

Let $F : Y \to Y'$ be relative Frobenius, let D' be the twist, and put $\mathcal{E} = F_*\mathcal{O}_Y(B_p)$. Smoothness makes F finite flat of degree p^3, so $\operatorname{rk}\mathcal{E} = p^3$. The tensor product A_* of the two local Frobenius-fibre algebras has dimension p^6. If t is the fibre coordinate and $t(z) = c \ne 0$, the full quotient is $A_*[t]/(t^p - c^p)$, free over A_* with basis $1, t, \dots, t^{p-1}$. Projection to the coefficient of 1 keeps exactly weights $j = pa$, with $0 \le a \le s$. Consequently the natural map

$$\bigoplus_{a+b=s} H^0(Y, B_p + paD) \otimes H^0(Y, B_p + pbD) \otimes \mathcal{O}_{Y'\times Y'}(-aD'_1 - bD'_2) \longrightarrow \mathcal{E} \boxtimes \mathcal{E}$$

spans A_* at the base pair. The projection formula identifies these summands with the evaluation map; the surviving fibre scalars c^{pa} are nonzero. The map therefore has generic rank p^6. No reducedness assumption on A_* is used. □

Lemma 2.20. *Fix the jet system, padding sections, canonical trivialization, flag, blowup, and complete-intersection witness over $\mathbb{C}$, with $\mu_{\min}(\Omega^1_X|_C) \ge 0$. They extend to a model whose smooth closed fibres occur in arbitrarily large characteristics and preserve the evaluation surjection, ampleness, intersection numbers, and semistability of $\Omega^1_Y|_C$ of slope zero.*

Proof. Choose the padding divisor T and the finitely many nonvanishing padding sections as in Proposition 2.19. Spread out the flag, canonical trivialization, jet sections, blowup,

birational model $f : V \to \widetilde{X}$, and ample divisors H_1, H_2. Shrinking the model preserves smoothness, ampleness, the finite jet surjection, and all intersection numbers. Since $\deg(\Omega^1_X|_C) = K_X \cdot C = 0$, the least-slope condition makes this curve bundle semistable of slope zero. Openness of semistability preserves it after further shrinking. There remain closed fibres in arbitrarily large characteristics. □

On each smooth fibre let $F : Y \to Y'$ denote relative Frobenius. For the following lemmas put

$$\ell_p = \lfloor p/k_0 \rfloor, \quad B_p = k_0 \ell_p P + T, \quad \mathcal{E}_p = F_* \mathcal{O}_Y(B_p), \quad s_p = p^3, \quad R_p = p^6.$$

All constants in $O(p^{-1})$ depend only on the fixed data.

2.2.2. *The diagonal rank.*

Lemma 2.21. *Let V be a semistable vector bundle of rank three and degree zero on a smooth projective curve of genus g in characteristic p. For $0 \le j \le 3(p-1)$, its truncated symmetric powers satisfy*

$$\mu_{\max}(\mathcal{A}_j(V)) \le 3c_g,$$

where c_g depends only on g and the rank, independently of j, p.

Proof. Langer's bound [10, Corollary 2.5] gives $L_{\min}(V) \ge -c_g/(p-1)$ with c_g depending only on the genus and rank. Put $u = 3(p-1)$. The perfect complementary degree pairing gives

$$\mathcal{A}_{u-j}(V) \simeq \mathcal{A}_j(V)^\vee \otimes (\det V)^{p-1}.$$

Truncated symmetric powers are quotients of tensor powers; their asymptotic least slopes are at least the corresponding multiple of $L_{\min}(V)$. Since $\deg \det V = 0$, the pairing implies $\mu_{\max}(\mathcal{A}_j(V)) \le 3c_g$, uniformly in j, p. □

Lemma 2.22. *Let $Y \supset S \supset C$ be a smooth complete intersection flag cut by successive ample Cartier divisors h_1, h_2. If $\mathcal{G}$ is locally free of rank e and $\mu_{\max}(\mathcal{G}|_C) \le M$, with $M \ge 0$, then*

$$h^0(Y, \mathcal{G}) \le e(M+1) \prod_{i=1}^{2} \left(1 + \frac{M}{h_i \cdot C}\right).$$

Proof. Successive restriction along the flag bounds sections by the sum of the spaces on C twisted by $-b_1 h_1 - b_2 h_2$, with $b_i \ge 0$. Only pairs satisfying $b_1 h_1 \cdot C + b_2 h_2 \cdot C \le M$ contribute, since negative maximal slope forbids sections. For each pair, evaluation at $\lfloor M \rfloor + 1$ points is injective and bounds the dimension by $e(M+1)$. There are at most $\prod_i (1 + M/(h_i \cdot C))$ pairs. The restriction sequences terminate because sufficiently negative ample twists have no sections. □

Lemma 2.23. *On a smooth fibre of Lemma 2.20, put* $B_p = k_0\lfloor p/k_0\rfloor P + T$ *and* $R_p = p^6$. *If* $(2P + sD)\cdot P^2 < 4P^3$ *and* $343P^3 < 1/4$, *the weighted evaluation map has rank less than* $R_p/4$ *on the diagonal of* $Y' \times Y'$.

Proof. Let $\Delta_F = Y \times_{Y'} Y$, and put

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + psD_1).$$

On Δ_F, the weighted section products all take values in this same line bundle, because $pD_1 = pD_2$ there. Their value rank at any point of the diagonal $Y' \subset Y' \times Y'$ is at most $h^0(\Delta_F, \mathcal{L}_p)$. The canonical Frobenius filtration [11, Corollary 3.5] has graded pieces

$$\mathcal{G}_j = \mathcal{O}_Y(2B_p + psD) \otimes \mathcal{A}_j(\Omega^1_Y), \quad 0 \le j \le 3(p-1), \qquad e_j = \operatorname{rk} \mathcal{A}_j, \quad \sum_j e_j = p^3,$$

Here $\mathcal{A}_j$ is the truncated symmetric power of degree j, locally the degree-j part of

$$\mathcal{O}[u_1, u_2, u_3]/(u_1^p, u_2^p, u_3^p).$$

By Lemma 2.21, the truncated powers on C have uniformly bounded maximal slopes. Set $\Lambda = l_1 l_2$, so $[C] = \Lambda P^2$ and $h_i \cdot C = l_i \Lambda P^3$. The differences

$$B_p - pP = T - (p - k_0\ell_p)P$$

belong to a fixed finite list. The displayed intersection hypothesis therefore gives one constant c_0, independent of j, p, for which

$$\mu_{\max}(\mathcal{G}_j|_C) \le M_p := 7p\Lambda P^3 + c_0.$$

Lemma 2.22 therefore gives

$$h^0(Y, \mathcal{G}_j) \le e_j p^3\big(343P^3 + O(p^{-1})\big).$$

Summing the graded pieces gives

$$h^0(\Delta_F, \mathcal{L}_p) \le R_p\big(343P^3 + O(p^{-1})\big) < R_p/4. \tag{3}$$

The last inequality follows from $343P^3 < 1/4$ for sufficiently large p. There is no extra factor for the number of graded pieces: their total rank is exactly p^3. □

2.2.3. *Determinant multiplicities.*

Lemma 2.24. *For the evaluation map in Proposition 2.19, put* $s_p = p^3$, $R_p = p^6$, *and* $\mathcal{E}_p = F_*\mathcal{O}_Y(B_p)$. *There exist line bundles*

$$\mathcal{Q}_i = (\det \mathcal{E}_p)^{s_p} \otimes \mathcal{O}_{Y'}(m_i D'), \qquad m_i \in \mathbb{Z}_{\ge 0}, \quad m_1 + m_2 = sR_p,$$

and a nonzero determinant section of $\mathcal{Q}_1 \boxtimes \mathcal{Q}_2$. *Under the numerical identification with* Y', *for* $\lambda_i = m_i/R_p$,

$$\frac{c_1(\mathcal{Q}_i)}{R_p} = P + \lambda_i D + O(p^{-1}), \qquad 0 \le \lambda_i \le s.$$

Proof. Proposition 2.19 gives generic evaluation rank R_p. Choose R_p generically independent evaluation columns of weights (a, b), $a + b = s$. Their determinant is a nonzero section

$$\sigma \in H^0(Y' \times Y', \mathcal{Q}_1 \boxtimes \mathcal{Q}_2), \qquad \mathcal{Q}_i = (\det \mathcal{E}_p)^{s_p} \otimes \mathcal{O}_{Y'}(m_i D'),$$

where $m_i \ge 0$ and $m_1 + m_2 = sR_p$. Let E_σ be the effective divisor of σ, so

$$E_\sigma \in |\mathcal{Q}_1 \boxtimes \mathcal{Q}_2|.$$

The Frobenius formula for the first Chern class [12, Lemma 4.2] gives, under the natural numerical identification with the twisted fibre,

$$\frac{c_1(\mathcal{Q}_i)}{R_p} = \frac{B_p}{p} + \frac{p-1}{2p} K_Y + \lambda_i D = P + \lambda_i D + O(p^{-1}), \qquad \lambda_i = m_i/R_p \in [0, s].$$

□

Lemma 2.25. *Let* L_1, L_2 *be line bundles on smooth projective varieties* Y_1, Y_2, *and let* $x_i \in Y_i$. *If every* $F_i \in |L_i|$ *satisfies* $\operatorname{mult}_{x_i}(F_i) \le a_i$, *then every* $F \in |L_1 \boxtimes L_2|$ *satisfies*

$$\operatorname{mult}_{(x_1,x_2)}(F) \le a_1 + a_2.$$

Proof. By Künneth the section space is the tensor product of the two section spaces. Choose bases adapted to their multiplicity filtrations. Their leading forms are independent in each degree, and tensor products remain independent in each bidegree. The largest possible nonzero initial degree is at most $a_1 + a_2$. □

Lemma 2.26. *Let* E_σ *be the determinant divisor in Lemma 2.24. The data* f, H_1, H_2, *specialized as in Lemma 2.20, give*

$$\operatorname{mult}_{(x',x')}(E_\sigma) \le R_p\big(2b + O(p^{-1})\big). \tag{4}$$

Proof. Twist the specialized blowup and its fixed movable witness as well. For every member $E_i \in |\mathcal{Q}_i|$, its proper transform is effective. Pulling it back to the specialized model V and intersecting with $H_1 \cdot H_2$ gives

$$\frac{\operatorname{mult}_{x'}(E_i)}{R_p} \le b + O(p^{-1}).$$

The fixed complete intersection retains positivity on the specialized model. The intersection inequality for effective divisors therefore applies to every member of these linear systems on the fibre.

Apply Lemma 2.25 to the two line bundles $\mathcal{Q}_i$. □

Lemma 2.27. *Let M be an $R \times R$ matrix of regular functions near a smooth point u, with* $\det M \neq 0$ *and* $\operatorname{rank} M(u) = r$. *Then*

$$\operatorname{mult}_u(\operatorname{div}(\det M)) \geq R - r.$$

Proof. Constant invertible row operations make $R-r$ rows vanish at u. Each determinant monomial contains one entry from each such row, so belongs to $\mathfrak{m}_u^{R-r}$. □

Proof of Lemma 2.16. Choose the flag from Lemma 2.17. Lemma 2.20 spreads the fixed flag, jet system, and curve class to smooth fibres in arbitrarily large characteristics p. Proposition 2.19 and Lemma 2.24 give a nonzero determinant divisor E_σ, with $R_p = p^6$. The diagonal rank bound in Lemma 2.23, the matrix bound in Lemma 2.27, and Lemma 2.26 give

$$\tfrac{3}{4}R_p < \operatorname{mult}_{(x',x')}(E_\sigma) \leq R_p\big(2b + O(p^{-1})\big). \tag{5}$$

Divide by R_p and let $p \to \infty$. Then $3/4 \leq 2b < 3/4$, a contradiction. □

Proof of Lemma 1.2. Fix $s > 0$ and assume $\kappa(D) = -\infty$. Take n sufficiently large for Lemma 2.1, $r_n > s$ and $343P_n^3 < 1/4$. Since $P_n - r_nD$ is nef,

$$(2P_n + sD) \cdot P_n^2 \leq (2 + s/r_n)P_n^3 < 4P_n^3.$$

Lemma 2.1 supplies a point x, its blowup $\beta : \widetilde{X} \to X$ with exceptional divisor E, and $f : V \to \widetilde{X}$, H_1, H_2 such that

$$f^*\beta^*(P_n + sD) \cdot H_1 \cdot H_2 < 40\epsilon f^*E \cdot H_1 \cdot H_2 < \tfrac{3}{8}f^*E \cdot H_1 \cdot H_2.$$

Apply Lemma 2.16 with $P = P_n$. Since $R(P_n, s) = R_{n,s}$, no $|kR_{n,s}|$ with $kR_{n,s}$ Cartier separates $8k$-jets at any $z \in U$. □

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